InPulse Academic Intelligence Platform – End to End Flow

Raw Academic Data → Analytics → Insights → Dashboards & Weekly HOD Reports → Email

1. SOURCE DATABASE – inpulse_db (Operational / Raw Data)

inpulse_db

Academic
- Students
- Departments
- Branches
- Batches / Sections
- Enrollments

Learning & Assessments
- Courses
- Modules
- Assessments
- Assessment Scores

Attendance
- Attendance Records

Feedback
- Feedback Forms
- Questions
- Responses

People
- Instructors
- Course Instructors Mapping

Source Principles
- Read Only for Analytics
- No changes to operational data
- Data validated before use

2. AIRFLOW PIPELINE – academic_intelligence_pipeline (Orchestrated Flow)

Apache Airflow

1. **Extract** – Extract required data from inpulse_db
2. **Data Quality Validation** – Validate schema, relationships, values, ranges, completeness
3. **Transform & Normalize** – Clean, standardize, aggregate and prepare data for analytics
4. **Compute Analytics** – Calculate KPIs for Student, Batch, Section, Course, Instructor, Dept
5. **Risk Classification** – Identify Academic, Attendance & High Risk Students
6. **Insight Generation** – Generate rule-based insights and recommendations

Airflow Features
- Scheduling
- Dependencies
- Retries
- Logging
- Alerting

3. ANALYTICS DATABASE – academic_analytics_db (Processed & Summarized Data)

academic_analytics_db

Student Performance Summary
- Academic %
- Attendance %
- Risk Category

Batch Performance Summary
- Batch KPIs
- Students Count
- Performance

Section Performance Summary
- Section KPIs
- Attendance
- Academic

Course Performance Summary
- Course KPIs
- Students
- Enrollment

Department Performance Summary
- Dept KPIs
- Students
- Courses

Feedback Summary
- Overall Rating
- Response Count
- Ratings Dist.

Module Feedback Summary
- Module Rating
- Response Count

Instructor Feedback Summary
- Instructor Rating
- Response Count

Risk Summary
- High Risk
- Academic Risk
- Attendance Risk

Insight Summary
- Insight
- Rule/Metric
- Recommendation
- Priority

Data Principles
- Analytics DB is separate
- Used for Reporting & Dashboards
- Contains only processed data

4A. POWER BI DASHBOARD (Interactive Analysis)

Power BI

1. Executive Overview (KPIs, Risk Distribution, Insights)
2. Department Performance (Dept KPIs, Course & Section Performance)
3. Student Risk Intelligence (Risk Distribution, Risk Students, Trends)
4. Course & Feedback Intelligence (Course KPIs, Module & Instructor Feedback)
5. Insights & Recommendations (Actionable Insights & Next Steps)

4B. WEEKLY REPORT GENERATION (Per Department)

1. **Prepare Data** – Filter data by Department & Reporting Period
2. **Generate Excel Report (.xlsx)**
   - Executive Summary
   - Batch Summary
   - Instructor Summary
   - Feedback Summary
   - Insights
   - Data Quality
3. **Generate PDF Report (.pdf)**
   - Executive Summary
   - Top Instructors
   - Low Performing Batches/Sections
   - Insights
   - Recommendations
   - Data Quality
4. **Save Reports** – /reports/weekly/YYYY-MM-DD/[department_code]/
   - dept_report.xlsx
   - dept_report.pdf

5. EMAIL DISTRIBUTION – Send Different Reports to Different HODs

1. **Get Department & HOD Details** – Read from departments table
   - department_id
   - department_code
   - department_name
   - hod_name
   - hod_email
2. **Compose Email**
   - Subject: Weekly Academic Report – {department_name}
   - Body: Please find attached the weekly academic report for your department.
3. **Attach Reports**
   - Attachments:
   - dept_report.pdf
   - dept_report.xlsx
4. **Send Email** – Using SMTP / Email Service
5. **Delivery Status** – Log Sent Status
   - Recipient (hod_email)
   - Report Files
   - Sent Time
   - Status: SENT
6. **On Failure** – Log Failure
   - Error Reason
   - Alert Admin
   - Retry (if configured)

6. SCHEDULER

Airflow DAG Trigger: Every Saturday (e.g., 08:00 AM)

DAG Flow: Extract → Validate → Transform → Analytics → Risk → Insights → Reports → Email → Logging

Timezone: Configurable

7. MONITORING & ALERTS

- Monitor DAG Runs
- Task Success / Failure
- Report Generation Status
- Email Delivery Status
- Alerts via Email / Slack on Failures

Outcome

Timely insights for better decision making, student support & academic improvement.

Legend: Source Layer · Processing Layer (Airflow) · Analytics Layer · Reporting Layer · Distribution Layer · Scheduling · Monitoring